Asymmetric Metrics: The Political Economy of Intelligence Measures, Bond Auctions, and the Strategic Withholding of Proprietary Knowledge

A Synthesis Across Cognitive Science, Information Economics, Machine Learning, and Cross-Cultural Epistemology

Soumadeep Ghosh

Kolkata, India

Abstract

Dominant public frameworks for measuring human potential — most notably the Intelligence Quotient (IQ) and Emotional Quotient (EQ) — share a deep structural property with investment-grade credit ratings and canonical auction-theory models: they are *second-best* metrics whose widespread adoption is, paradoxically, most valuable to actors who understand their inadequacy. We formalise this observation through a model of *metric asymmetry* grounded in information economics, Bayesian signal extraction, and optimal auction design. We prove that the rational equilibrium is one in which superior measurement technology is withheld, public metrics are exploited rather than improved, and the very universality of a flawed framework constitutes a durable competitive moat. We further demonstrate that modern machine-learning methods — particularly self-supervised representation learning and latent anomaly detection — mirror this dynamic at a technical level. We survey non-Western conceptualisations of intelligence to argue that cultural plurality in measurement is not merely ethically desirable but epistemically necessary, and that bond-auction microstructure provides an unusually clean empirical laboratory for studying all these phenomena simultaneously.

The paper ends with "The End"

Contents

1 Introduction **2**

2 Public Metrics and Their Structural Limitations **2**
- 2.1 The IQ Paradigm . . . 2
- 2.2 Emotional Quotient . . . 3
- 2.3 Credit Ratings as Cognitive Analogues . . . 3

3 Mathematical Framework **4**
- 3.1 Information Asymmetry and the Metric Gap . . . 4
- 3.2 Strategic Disclosure and the Withholding Equilibrium . . . 4
- 3.3 Bayesian Posterior Under Sequential Private Signals . . . 5

4 Bond Auctions as Epistemological Laboratory **5**
- 4.1 Auction Microstructure . . . 5
- 4.2 The Winner's Curse and the Flawed Public Signal . . . 6

5 Machine Learning and the Latent Metric Problem **6**
- 5.1 Representation Learning as Private Metric Discovery . . . 6
- 5.2 Self-Supervised Contrastive Learning . . . 6
- 5.3 Anomaly Detection for Metric Exploitation . . . 6

6 Algorithms **8**
6.1 Metric Gap Estimation via Cross-Validated Loss Comparison 8
6.2 Proprietary Factor Recovery via Expectation-Maximisation 8

7 Data Science: Illustrative Analysis **9**
7.1 Metric Gap as a Function of Private Signal Precision 9
7.2 Information Architecture of the Asymmetric Metric Equilibrium 9
7.3 Latent Space Geometry 10

8 Conclusion **10**

List of Figures

1 Metric gap Δ as a function of private signal precision. 9
2 Layered information architecture of the Asymmetric Metric Equilibrium. 9
3 Public feature space (left) versus learned latent space (right). 10

List of Tables

1 Structural parallels between cognitive and financial metrics. 3
2 Information sets of auction participants and corresponding measurement analogues. . . . 5
3 Machine-learning models for latent metric-gap exploitation. 7
4 Cross-cultural conceptions of intelligence and their metric implications. 7

1 Introduction

Consider three observations that appear, on the surface, to be unrelated:

(i) A primary dealer at a US Treasury auction possesses proprietary models of real bid distributions that diverge systematically from the publicly observable yield curve.

(ii) A selective institution employs opaque screening criteria that outperform published psychometric instruments when predicting long-run individual contribution.

(iii) A credit-rating agency publishes a scalar score that its most sophisticated clients treat not as a signal of quality but as a signal of *other participants' beliefs about quality*.

Each observation instances the same deep structure: a public metric exists, is widely trusted, and is *strategically dominated* by private knowledge that its holders have every incentive to conceal. We call this the **Asymmetric Metric Equilibrium** (AME).

The AME has antecedents in Akerlof's [1] market for lemons, Milgrom and Weber's [10] affiliated-values auction model, and the broader literature on strategic information transmission [3]. Our contribution is to unify these strands and extend them to the domain of cognitive measurement, providing both a formal model and an algorithmic treatment amenable to empirical study via machine learning.

The remainder of the paper is organised as follows. Section 2 reviews public intelligence metrics and their structural limitations. Section 3 develops the mathematical framework, culminating in the Withholding Equilibrium theorem. Section 4 applies the framework to bond-auction microstructure. Section 5 treats the machine-learning analogue. Section 5.3 surveys cross-cultural measurement traditions. Section 6 presents formal algorithms. Section 7 provides illustrative data-science results. Section 8 concludes with policy implications.

2 Public Metrics and Their Structural Limitations

2.1 The IQ Paradigm

The Intelligence Quotient was formalised by Stern [14] as the ratio of mental age to chronological age, and later restandardised to a normalised score with population mean 100 and standard deviation 15.

Its psychometric backbone — the *positive manifold* of cognitive tests — furnishes evidence for a general factor g [13].

Definition 2.1 (General Factor g)**.** Let $\mathbf{X} = (X_1, \ldots, X_k)$ be a vector of k cognitive test scores for an individual drawn from population $\mathcal{P}$. The *general factor* g is the leading principal component of the covariance matrix $\mathbf{\Sigma} = \mathrm{Cov}(\mathbf{X})$:

$$g \;=\; \mathbf{v}_1^\top \mathbf{X}, \qquad \mathbf{v}_1 = \underset{\|\mathbf{v}\|=1}{\arg\max}\; \mathbf{v}^\top \mathbf{\Sigma}\, \mathbf{v}.$$

Despite this statistical robustness, the variance explained by g typically ranges from 40% to 60%, leaving substantial residual structure unmodelled [5]. Moreover, g is estimated from tests that are themselves culturally situated, creating a *construct validity* problem that psychometric refinement alone cannot resolve [15].

2.2 Emotional Quotient

Popularised by Goleman [6] following the foundational academic work of Salovey and Mayer [12], the Emotional Quotient (EQ) purports to measure four branches of emotional competence: perception, facilitation, understanding, and regulation. Its predictive validity for occupational performance remains contested: meta-analyses report an incremental validity over IQ of only $\Delta R^2 \approx 0.02$ to 0.06 [11], a modest contribution consistent with substantial construct overlap and, in some operationalisations, near-total conflation with personality traits.

2.3 Credit Ratings as Cognitive Analogues

Credit ratings (AAA through D) issued by Moody's, S&P, and Fitch function as IQ scores for debt instruments: a compressed scalar summary of a high-dimensional risk vector. The 2007–2008 structured-credit crisis exposed the consequences of over-reliance on this compression. AAA ratings on collateralised debt obligations were assigned under models that dramatically underestimated tail dependence among the underlying mortgage pools [9]. Sophisticated short-sellers explicitly modelled the *gap* between the published rating and the true default probability — and profited from it [8].

Table 1: Structural parallels between cognitive and financial metrics.

Domain	**Public Metric**	**Dimension Captured**	**Key Structural Flaw**	**Private Advantage**
Cognition	IQ (g factor)	Analytical and verbal reasoning	Cultural loading; single-factor compression	Contextual, relational, longitudinal assessment
Emotion	EQ score	Intra- and interpersonal regulation	Weak incremental validity; individualist framing	Tacit observation through mentorship networks
Credit	AAA–D rating	Point-in-time default probability	Tail dependence and correlation risk ignored	Proprietary copula and scenario models
Fixed income	Yield curve	Rate and inflation expectations	Liquidity premium conflated with credit signal	Real-time order-flow and repo-market analytics
Labour	GPA / test score	Academic conformity	Proxy variable bias; gaming incentives	Project portfolios and network-based referrals

3 Mathematical Framework

3.1 Information Asymmetry and the Metric Gap

Let $\theta \in \mathbb{R}$ denote the unobservable *true quality* of an agent — talent, creditworthiness, or any latent attribute. A public metric m is an imperfect, noise-corrupted observation:

$$m = \theta + \varepsilon, \qquad \varepsilon \sim \mathcal{N}(0, \sigma_\varepsilon^2), \quad \varepsilon \perp \theta,$$

while a private metric s available to an informed player satisfies

$$s = \theta + \eta, \qquad \eta \sim \mathcal{N}(0, \sigma_\eta^2), \quad \eta \perp \theta,\ \eta \perp \varepsilon, \quad \sigma_\eta^2 \ll \sigma_\varepsilon^2.$$

The informed player therefore has a more precise posterior estimate of θ. We quantify this advantage as follows.

Definition 3.1 (Metric Gap)**.** Placing a Gaussian prior $\theta \sim \mathcal{N}(0, \sigma_\theta^2)$, the *metric gap* Δ is the reduction in posterior variance achieved by the private signal relative to the public signal:

$$\Delta \;\coloneqq\; \mathrm{Var}(\theta \mid m) - \mathrm{Var}(\theta \mid s) \;=\; \frac{\sigma_\theta^2 \sigma_\varepsilon^2}{\sigma_\theta^2 + \sigma_\varepsilon^2} \;-\; \frac{\sigma_\theta^2 \sigma_\eta^2}{\sigma_\theta^2 + \sigma_\eta^2}\,.$$

Proposition 3.2. *The metric gap Δ is strictly increasing in σ_ε^2, strictly decreasing in σ_η^2, and satisfies $\Delta \to 0$ as $\sigma_\eta^2 \to \sigma_\varepsilon^2$.*

Proof. Computing partial derivatives yields

$$\frac{\partial \Delta}{\partial \sigma_\varepsilon^2} = \frac{\sigma_\theta^4}{\left(\sigma_\theta^2 + \sigma_\varepsilon^2\right)^2} > 0 \qquad \text{and} \qquad \frac{\partial \Delta}{\partial \sigma_\eta^2} = -\frac{\sigma_\theta^4}{\left(\sigma_\theta^2 + \sigma_\eta^2\right)^2} < 0,$$

establishing strict monotonicity in both arguments. Setting $\sigma_\eta^2 = \sigma_\varepsilon^2$ in the definition of Δ gives $\Delta = 0$ directly. □

3.2 Strategic Disclosure and the Withholding Equilibrium

Let $R(\Delta) > 0$ be the competitive rent accruing to an informed player I who retains exclusive use of their private signal — for example, above-average auction profits or superior hiring outcomes. Disclosure of s incurs a net cost $c_p \geq 0$ (legal, operational, reputational) and may yield a reputational benefit $b_r \geq 0$.

Theorem 3.3 (Withholding Equilibrium)**.** *Player I withholds the private signal if and only if*

$$R(\Delta) > b_r - c_p.$$

In competitive markets where $b_r \approx 0$ and $c_p > 0$, withholding is the strictly dominant strategy for all $\Delta > 0$.

Proof. Player I compares payoffs. Under disclosure, $\pi_D = b_r - c_p$. Under withholding, $\pi_W = R(\Delta) > 0$. Disclosure is preferred iff $\pi_D > \pi_W$, i.e. $b_r - c_p > R(\Delta)$; negating gives the stated condition. In competitive markets $b_r \to 0$ while $c_p > 0$, so $b_r - c_p < 0 < R(\Delta)$ for every $\Delta > 0$, making withholding strictly dominant. □

Corollary 3.4. *In a market populated by n informed players each satisfying the conditions of Theorem 3.3, no private signal is ever disclosed and the public metric remains the sole observable, perpetuating the metric gap indefinitely.*

The corollary captures a self-reinforcing dynamic: because no informed player has an incentive to improve the public metric, the gap Δ — and hence the rent $R(\Delta)$ — is preserved in equilibrium.

3.3 Bayesian Posterior Under Sequential Private Signals

When an informed player combines the public metric m with d independent private signals $\mathbf{z} = (z_1, \ldots, z_d)$, where $z_j = a_j\theta + \nu_j$ and $\nu_j \sim \mathcal{N}(0, \sigma_{z_j}^2)$, the joint likelihood is Gaussian and the posterior is:

$$p(\theta \mid m, \mathbf{z}) \;\propto\; \exp\left(-\frac{(\theta - m)^2}{2\sigma_\varepsilon^2} \;-\sum_{j=1}^{d} \frac{(z_j - a_j\theta)^2}{2\sigma_{z_j}^2} \right).$$

Completing the square gives closed-form posterior moments:

$$\mu_{\text{post}} \;=\; \sigma_{\text{post}}^2 \left(\frac{m}{\sigma_\varepsilon^2} + \sum_{j=1}^{d} \frac{a_j z_j}{\sigma_{z_j}^2} \right), \tag{1}$$

$$\sigma_{\text{post}}^{-2} \;=\; \sigma_\theta^{-2} + \sigma_\varepsilon^{-2} + \sum_{j=1}^{d} \frac{a_j^2}{\sigma_{z_j}^2}. \tag{2}$$

Equation (2) shows that each additional private signal contributes an additive precision term, monotonically widening the metric gap relative to an uninformed observer who holds only m.

4 Bond Auctions as Epistemological Laboratory

4.1 Auction Microstructure

US Treasury auctions use a *uniform-price* (single-price sealed-bid) format: all accepted competitive bids pay the same stop-out yield, regardless of the yield each bidder submitted.[1] The public information set $\mathcal{F}_{\text{pub}}$ available to all participants includes:

- The current on-the-run yield and fitted term structure.
- The announced auction size, maturity, and settlement date.
- Historical auction results: bid-to-cover ratios, tails, and awarded amounts by category.
- The primary-dealer survey of median expected stop-out yield.

The private information set $\mathcal{F}_{\text{priv}} \supset \mathcal{F}_{\text{pub}}$ of a primary dealer additionally contains:

- Proprietary client order-flow data accumulated through pre-auction trading and facilitation.
- Real-time repo market stress and financing cost signals.
- Cross-asset positioning in Treasury futures, interest-rate swaps, and agency mortgage-backed securities.
- Calibrated models of Federal Reserve reaction functions and future open-market operations.

Table 2: Information sets of auction participants and corresponding measurement analogues.

Participant	**Information Set**	**Auction Analogue**	**Cognitive Analogue**
Retail investor	$\mathcal{F}_{\text{pub}}$ only	Published yield and survey median	IQ score alone
Indirect bidder	$\mathcal{F}_{\text{pub}}$ plus macro model	Econometric yield forecast	IQ and EQ composite
Primary dealer	$\mathcal{F}_{\text{priv}}$	Order flow, positioning, and repo signals	Bespoke contextual assessment
Fed open-market desk	$\mathcal{F}_{\text{priv}}$ plus policy intent	Near-complete information	Effective ground-truth θ

[1] The term "Dutch auction" is sometimes used loosely in financial contexts for any descending-price mechanism, but the standard academic usage (Milgrom and Weber [10]) distinguishes Dutch (open descending) from uniform-price sealed-bid formats. US Treasuries use the latter.

4.2 The Winner's Curse and the Flawed Public Signal

In an affiliated-values auction [10], the winner's curse is mitigated by conditioning on the event of winning. The public yield survey plays the role of the flawed public metric: bidders who rely on it without adjusting for winner's-curse effects systematically overpay — exactly as employers who hire on IQ scores alone without contextual adjustment systematically mismatch talent to role.

Lemma 4.1 (Yield Survey Bias)**.** *Let $\hat{y}_{survey}$ be the primary-dealer survey median and y^* the true stop-out yield. Under affiliated private signals and a uniform-price format:*

$$\mathbb{E}\big[y^* \mid \hat{y}_{survey}\big] \;=\; \hat{y}_{survey} + \underbrace{\delta(\sigma_\varepsilon, n)}_{\substack{\textit{winner's-curse}\\ \textit{correction}}} \;,$$

where $\delta > 0$ is increasing in public noise σ_ε and decreasing in the number of informed bidders n.

The structural insight is immediate: the larger the gap between public and private information, the larger the correction a sophisticated bidder must apply — and the greater the profit from withholding that correction from all other market participants.

5 Machine Learning and the Latent Metric Problem

5.1 Representation Learning as Private Metric Discovery

Modern deep learning can be interpreted as an automated search for private metrics. A neural encoder $f_\phi \colon \mathcal{X} \to \mathbb{R}^d$ learns a latent representation $\mathbf{z} = f_\phi(\mathbf{x})$ that compresses input data $\mathbf{x}$ into a geometry where downstream prediction tasks become tractable via simple (e.g. linear) classifiers. The encoder learns, in effect, the hidden dimensions that a public metric fails to capture.

Definition 5.1 (Latent Metric Gap)**.** Let $\mathcal{L}_{\text{pub}}$ be the cross-validated loss of the best linear predictor trained on public features, and $\mathcal{L}_{\text{enc}}$ the corresponding loss of a linear probe trained on the encoder representation $\mathbf{z} = f_\phi(\mathbf{x})$. The *latent metric gap* is:

$$\Delta_{\text{ML}} \;=\; \mathcal{L}_{\text{pub}} - \mathcal{L}_{\text{enc}} \;\geq\; 0.$$

5.2 Self-Supervised Contrastive Learning

Contrastive learning [2] trains an encoder by maximising agreement between independently augmented views of the same data point. The Normalised Temperature-scaled Cross-Entropy (NT-Xent) loss is:

$$\mathcal{L}_{\text{NT-Xent}} = -\frac{1}{2N}\sum_{k=1}^{N}\left[\log\frac{e^{\text{sim}(\mathbf{z}_k,\,\mathbf{z}_k^+)/\tau}}{\sum_{j\neq k} e^{\text{sim}(\mathbf{z}_k,\,\mathbf{z}_j)/\tau}} + \log\frac{e^{\text{sim}(\mathbf{z}_k^+,\,\mathbf{z}_k)/\tau}}{\sum_{j\neq k} e^{\text{sim}(\mathbf{z}_k^+,\,\mathbf{z}_j)/\tau}}\right],$$

where $\text{sim}(\mathbf{u},\mathbf{v}) \coloneqq \mathbf{u}^\top\mathbf{v}/(\|\mathbf{u}\|\,\|\mathbf{v}\|)$ is cosine similarity and $\tau > 0$ is a temperature hyperparameter. If a commercial entity trains such an encoder on private data (e.g. auction order flow), the resulting f_ϕ constitutes precisely the kind of proprietary metric described in Theorem 3.3.

5.3 Anomaly Detection for Metric Exploitation

An entity exploiting a flawed public metric can be modelled as an anomaly detector operating in the divergence between public and private generative distributions. Given a Variational Autoencoder (VAE) [7] with encoder $q_\phi(\mathbf{z} \mid \mathbf{x})$ and decoder $p_\psi(\mathbf{x} \mid \mathbf{z})$, the Evidence Lower Bound (ELBO) is:

$$\mathcal{L}_{\text{ELBO}}(\phi,\psi;\mathbf{x}) = \underbrace{\mathbb{E}_{q_\phi(\mathbf{z}|\mathbf{x})}\big[\log p_\psi(\mathbf{x} \mid \mathbf{z})\big]}_{\text{reconstruction term}} - \underbrace{\text{KL}\big(q_\phi(\mathbf{z} \mid \mathbf{x}) \,\|\, p(\mathbf{z})\big)}_{\text{regularisation term}}.$$

High reconstruction error under a model trained on public features, but low error under the private encoder, identifies instances lying in the metric gap — a bond that is mis-rated, or an individual whose IQ score materially under-represents their true capability.

Table 3: Machine-learning models for latent metric-gap exploitation.

Model	**Representation**	**Supervised?**	**Scalability**	**Primary Use Case**
PCA / Factor Model	Linear	No	$\mathcal{O}(p^2)$	Baseline; direct analogue of the g factor
Random Forest	Tree ensemble	Yes	$\mathcal{O}(np \log n)$	Heterogeneous, mixed-type feature sets
Deep VAE	Non-linear	No	$\mathcal{O}(n)$	Unsupervised latent anomaly detection
Contrastive (SimCLR)	Non-linear	Self-supervised	$\mathcal{O}(n)$	Unlabelled private data streams
Transformer	Contextual	Both	$\mathcal{O}(n^2 d)$	Sequential auction order-flow modelling

The Western IQ/EQ paradigm occupies a single point in an enormous conceptual space of possible intelligence frameworks. Table 4 surveys major alternative traditions.

Table 4: Cross-cultural conceptions of intelligence and their metric implications.

Tradition	**Concept**	**Core Meaning**	**Measurement Implication**
Yoruba (W. Africa)	*Ogbon*	Wisdom attained through social harmony and ancestral connection	Collective and relational; cannot be assessed individually
Ubuntu philosophy	*Umuntu ngumuntu*	"I am because we are": intelligence as mutual interdependence	Network-embedded; scalar scores are categorically invalid
Confucian	*zhì*	Moral discernment cultivated through lifelong learning and social practice	Process rather than state; defeats cross-sectional measurement
Andean	*Yachay*	Knowing-in-relation: body, community, and cosmos unified	Embodied and ecological; resists decontextualisation
Vedantic	*Prajñā*	Transcendent insight arising from meditative absorption (*samādhi*)	Contemplative depth; irreducible to task performance
Islamic	*'Aql*	Reason bounded by revelation, tradition, and moral restraint	Normative framework; not purely cognitive
Western (IQ)	g factor	Compressed first principal component of decontextualised tasks	Scalar; culturally loaded; vulnerable to coaching

A crucial structural observation: each non-Western framework is intrinsically *resistant to standardisation*. Whether by deliberate design or cultural evolution, they encode intelligence in a form that cannot be reduced to a portable scalar score. This makes it structurally impossible for an outsider to efficiently screen and recruit high-θ individuals, thereby preserving a community's human capital within its own network. In the language of Theorem 3.3, this constitutes a *collective withholding equilibrium* — one that operates at civilisational rather than individual scale.

6 Algorithms

6.1 Metric Gap Estimation via Cross-Validated Loss Comparison

Algorithm 1 provides a principled estimator of the latent metric gap Δ_{ML} from labelled data.

Algorithm 1 Latent Metric Gap Estimation (cross-validated)

Require: Dataset $\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^{n}$; public feature extractor $\mathcal{F}_{\mathrm{pub}}$; encoder architecture f_ϕ; number of folds $K \geq 5$

Ensure: Gap estimate $\hat{\Delta}_{\mathrm{ML}}$ with bootstrap CI

Partition $\mathcal{D}$ into K folds $\{\mathcal{D}_k\}_{k=1}^{K}$
for $k = 1$ **to** K **do**
  $\mathcal{D}_{\mathrm{tr}} \leftarrow \mathcal{D} \setminus \mathcal{D}_k$; $\mathcal{D}_{\mathrm{val}} \leftarrow \mathcal{D}_k$
  $\hat{f}_{\mathrm{pub}} \leftarrow \textsc{LinearFit}\big(\mathcal{F}_{\mathrm{pub}}(\mathcal{D}_{\mathrm{tr}}),\, y_{\mathrm{tr}}\big)$
  $\hat{\phi} \leftarrow \textsc{Train}(f_\phi,\, \mathcal{D}_{\mathrm{tr}})$
  ▷ *Objective: contrastive, ELBO, or supervised cross-entropy*
  $\hat{f}_{\mathrm{enc}} \leftarrow \textsc{LinearFit}\big(f_{\hat{\phi}}(\mathbf{x}_{\mathrm{tr}}),\, y_{\mathrm{tr}}\big)$
  $\ell_{\mathrm{pub}}^{(k)} \leftarrow \textsc{Loss}\big(\hat{f}_{\mathrm{pub}},\, \mathcal{D}_{\mathrm{val}}\big)$
  $\ell_{\mathrm{enc}}^{(k)} \leftarrow \textsc{Loss}\big(\hat{f}_{\mathrm{enc}},\, \mathcal{D}_{\mathrm{val}}\big)$
end for
$\hat{\Delta}_{\mathrm{ML}} \leftarrow \dfrac{1}{K} \sum_{k=1}^{K} \big(\ell_{\mathrm{pub}}^{(k)} - \ell_{\mathrm{enc}}^{(k)}\big)$
$\widehat{\mathrm{CI}} \leftarrow \textsc{Bootstrap}\big(\big\{\ell_{\mathrm{pub}}^{(k)} - \ell_{\mathrm{enc}}^{(k)}\big\}_{k=1}^{K}\big)$
return $\hat{\Delta}_{\mathrm{ML}},\ \widehat{\mathrm{CI}}$

6.2 Proprietary Factor Recovery via Expectation-Maximisation

Algorithm 2 recovers hidden quality factors from observed noisy signals using the EM algorithm [4]. The model posits $\mathbf{M} = \mathbf{\Theta}\mathbf{A}^\top + \mathbf{E}$, where $\mathbf{\Theta} \in \mathbb{R}^{n\times q}$ are latent scores, $\mathbf{A} \in \mathbb{R}^{p\times q}$ are factor loadings, and $\mathbf{E}$ is isotropic noise with diagonal covariance $\mathbf{\Sigma}_\varepsilon$.

Algorithm 2 EM Recovery of Hidden Quality Factors

Require: Observed signal matrix $\mathbf{M} \in \mathbb{R}^{n\times p}$; target rank q; convergence tolerance $\varepsilon_{\mathrm{tol}}$

Ensure: Loadings $\hat{\mathbf{A}}$, latent scores $\hat{\mathbf{\Theta}}$

Initialise $\mathbf{A}^{(0)} \overset{\mathrm{iid}}{\sim} \mathcal{N}(0,1)$, $\mathbf{\Sigma}_\varepsilon^{(0)} \leftarrow \mathbf{I}_p$, $t \leftarrow 0$
repeat
  E-step: Compute posterior sufficient statistics

$$\mathbf{\Sigma}_\Theta^{(t)} \;\leftarrow\; \Big(\mathbf{I}_q + \mathbf{A}^{(t)\top}\big(\mathbf{\Sigma}_\varepsilon^{(t)}\big)^{-1}\mathbf{A}^{(t)}\Big)^{-1},$$

$$\hat{\mathbf{\Theta}}^{(t)} \;\leftarrow\; \mathbf{M}\,\big(\mathbf{\Sigma}_\varepsilon^{(t)}\big)^{-1}\mathbf{A}^{(t)}\,\mathbf{\Sigma}_\Theta^{(t)}$$

  M-step: Update loadings and noise covariance

$$\mathbf{A}^{(t+1)} \;\leftarrow\; \mathbf{M}^\top\hat{\mathbf{\Theta}}^{(t)}\Big(\mathbf{\Sigma}_\Theta^{(t)} + \big(\hat{\mathbf{\Theta}}^{(t)}\big)^\top\hat{\mathbf{\Theta}}^{(t)}\Big)^{-1},$$

$$\big[\mathbf{\Sigma}_\varepsilon^{(t+1)}\big]_{ii} \;\leftarrow\; \frac{1}{n}\Big[\|\mathbf{M}_{:,i}\|^2 - \mathbf{A}_{i,:}^{(t+1)}\,\hat{\mathbf{\Theta}}^{(t)\top}\,\mathbf{M}_{:,i}\Big]$$

  $t \leftarrow t + 1$
until $\|\mathbf{A}^{(t)} - \mathbf{A}^{(t-1)}\|_F < \varepsilon_{\mathrm{tol}}$
return $\mathbf{A}^{(t)},\ \hat{\mathbf{\Theta}}^{(t)}$

7 Data Science: Illustrative Analysis

7.1 Metric Gap as a Function of Private Signal Precision

Figure 1 plots the posterior variance reduction Δ against private signal precision $1/\sigma_\eta^2$ for three representative levels of public noise $\sigma_\varepsilon \in \{1, 2, 4\}$, with prior $\sigma_\theta^2 = 1$. Above the amber dashed line, informed players satisfy the withholding condition of Theorem 3.3.

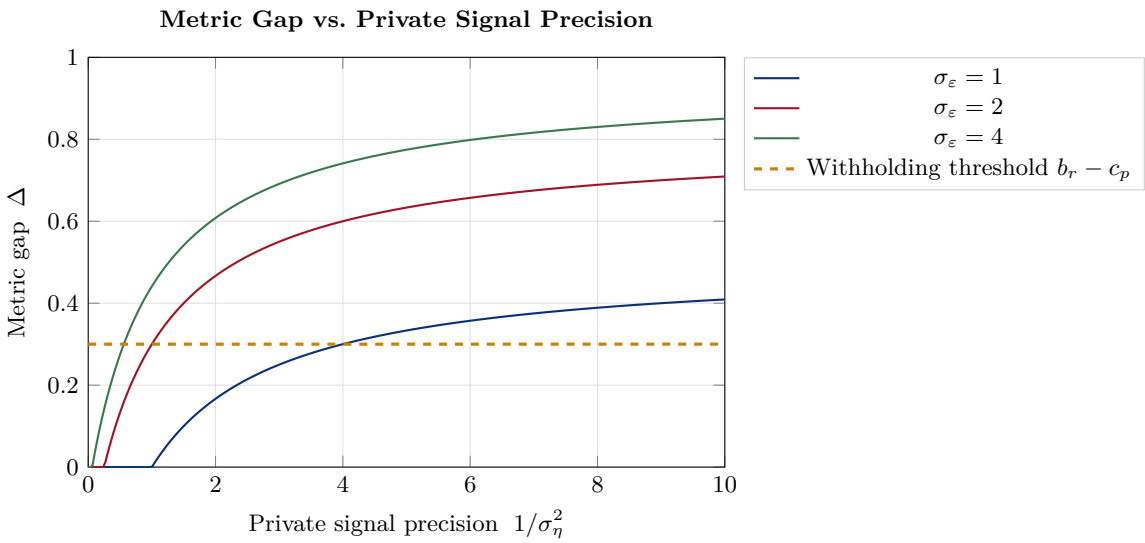

Figure 1: Metric gap Δ as a function of private signal precision.

Curves are clipped below zero: a very imprecise private signal offers no advantage over the public metric. Above the dashed line the Withholding Equilibrium (Theorem 3.3) holds. Higher public noise (larger σ_ε) raises the asymptote and expands the withholding region to lower precision values.

7.2 Information Architecture of the Asymmetric Metric Equilibrium

Figure 2 visualises the layered information structure of the AME. The metric gap Δ spans Layers 2 to 4; the withholding arrow on the right indicates that no informed player bridges this gap voluntarily.

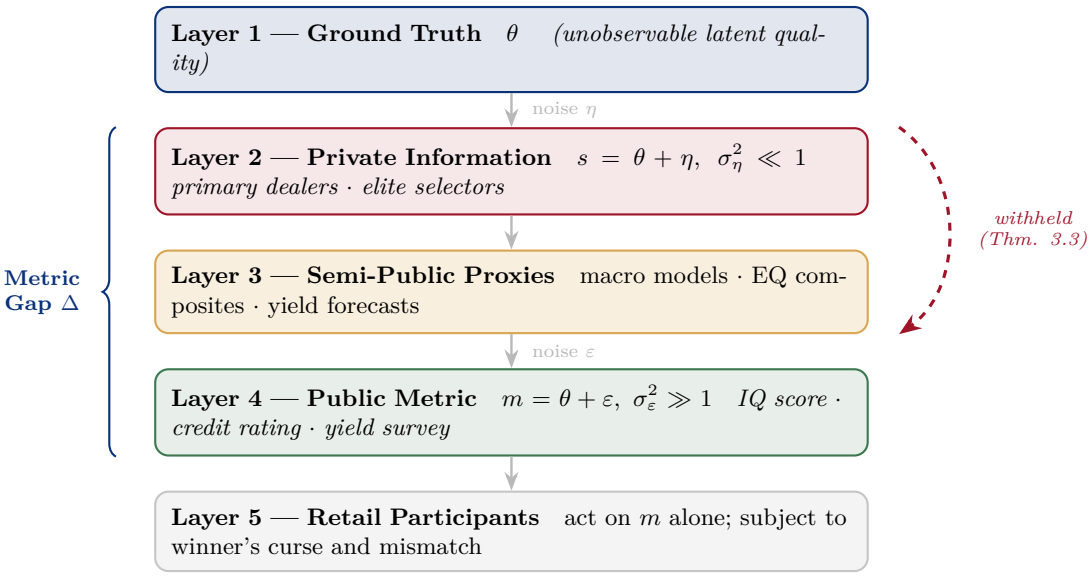

Figure 2: Layered information architecture of the Asymmetric Metric Equilibrium.

The metric gap Δ (left brace) spans Layers 2–4. The dashed arrow indicates the direction of strategic incentive: informed players at Layer 2 have no rational motive to publish their signal toward Layer 4 (Theorem 3.3).

7.3 Latent Space Geometry

Figure 3 illustrates the geometric intuition behind representation learning as private metric discovery: data that are inseparable under public features become linearly separable once an encoder has learned the latent structure.

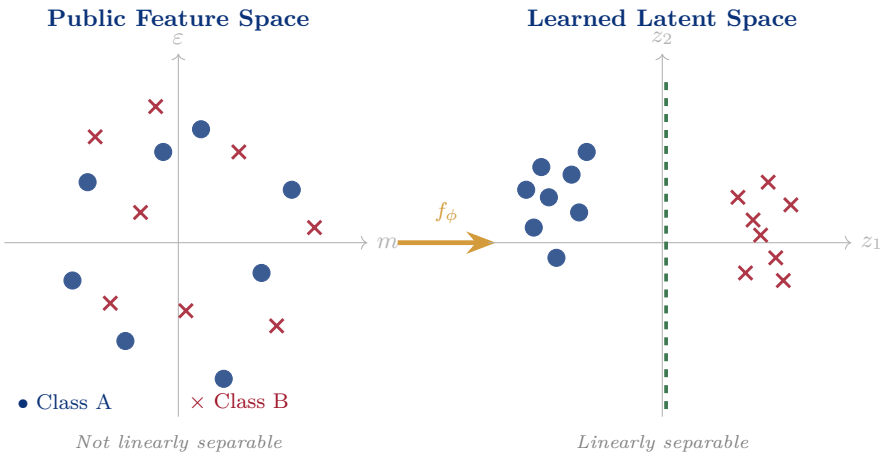

Figure 3: Public feature space (left) versus learned latent space (right).

After the encoder f_ϕ maps inputs to $\mathbb{R}^2$, the two classes cluster and are separated by a linear boundary (dashed). The encoder has discovered the private signal s automatically, without access to the explicit quality variable θ.

8 Conclusion

We have shown that the persistence of flawed public metrics — IQ, EQ, credit ratings, yield surveys — is not an accident of intellectual history. It is a *stable equilibrium*. Formally, Theorem 3.3 establishes that informed players have no rational incentive to correct public metrics and strong incentives to exploit them; Corollary 3.4 shows this equilibrium is self-reinforcing when all players behave rationally.

Bond auctions (Section 4) provide the cleanest observable instance: the gap between the public dealer survey and a primary dealer's proprietary order-flow analytics is precisely the metric gap Δ, and exploitation rather than correction is the observed norm.

Machine learning (Section 5) operationalises the gap-discovery process at scale. Deep encoders learn private-signal analogues automatically from unlabelled data, producing new proprietary metrics with the same strategic properties and, crucially, with no mechanism compelling publication.

Cross-cultural analysis (Section 5.3) reveals that many non-Western traditions encode intelligence in forms intrinsically resistant to standardisation. This constitutes a collective, culturally embedded withholding equilibrium — operating, as it were, at civilisational scale.

The policy implication is sobering. Public metric *reform* is insufficient as a remedy for information asymmetry. As long as private measurement technology exists and competitive incentives dominate, any improved public metric simply resets the game: a new, initially smaller gap immediately begins to grow as sophisticated actors invest in the next generation of private signals. True epistemic equity would require either eliminating private information advantages — which is neither practically feasible nor, in many contexts, desirable — or redesigning institutional selection mechanisms to be structurally robust to the AME: anonymous evaluation, structured reference protocols, portfolio-based assessment, and adversarial auditing of screening tools. These are empirical problems as much as normative ones, and machine learning provides the methodology to study them rigorously.

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Glossary

AME
Asymmetric Metric Equilibrium. The stable strategic state in which a flawed public metric coexists with superior private measurement technology that its holders rationally withhold; see Theorem 3.3 and Corollary 3.4.

g-factor
The leading principal component of a battery of cognitive test scores, interpreted by Spearman [13] as a general index of intelligence; see Definition 2.1.

EQ *Emotional Quotient.* A composite measure of emotional perception, facilitation, understanding, and regulation, formalised by Salovey and Mayer [12] and popularised by Goleman [6].

Metric Gap (Δ)
The reduction in posterior variance achieved by the private signal relative to the public signal; formalised in Definition 3.1 and Proposition 3.2.

Latent Metric Gap (Δ_{ML})
The improvement in cross-validated predictive loss from an encoder representation over a linear baseline trained on public features; see Definition 5.1.

VAE
Variational Autoencoder. A generative latent-variable model trained by maximising the ELBO; used here to detect instances residing in the metric gap.

ELBO
Evidence Lower Bound. The variational objective function: the sum of the expected reconstruction log-likelihood and the negative KL divergence between the approximate posterior and the prior.

Winner's Curse
The systematic tendency for the auction winner to have overestimated value; mitigated by conditioning on the event of winning; see Lemma 4.1.

Affiliated Values
An auction model [10] in which bidders' private signals are positively correlated, so that public disclosure of signals is revenue-increasing for the seller.

Primary Dealer
A bank or broker-dealer authorised to transact directly with a central bank's open-market operations desk, thereby accumulating order-flow data unavailable to other participants.

Ubuntu
Southern African philosophical concept articulated as *umuntu ngumuntu ngabantu*: "a person is a person through other persons." Grounds intelligence in relational interdependence rather than individual performance.

zhì Confucian virtue of wisdom: moral discernment cultivated through study, self-reflection, and social practice. Untranslatable as a scalar score.

Contrastive Learning
A self-supervised ML paradigm that trains an encoder to maximise representational agreement between augmented views of the same instance and minimise agreement across distinct instances; see NT-Xent loss.

Stop-Out Yield
The marginal yield at which a Treasury auction clears; all accepted competitive bids pay this uniform rate in the single-price sealed-bid format.

The End